

AZURE & MSSQL DBA MASTERCLASS

Master Database Administration from Beginner to Enterprise Level

Become a highly skilled Microsoft SQL Server & Azure Database Administrator by mastering on-premises SQL Server, Azure SQL services, security, performance tuning, high availability, disaster recovery, automation, and cloud migration through hands-on projects.

Course Roadmap (Beginner → Advanced)

01. Database Fundamentals

- Introduction to Databases & RDBMS
 - DBMS vs RDBMS
 - SQL Server Architecture
 - SQL Server Editions & Licensing
 - Database Objects
 - Data Types
 - CRUD Operations
 - Basic SQL Queries
 - Joins, Views & Constraints
-

02. Microsoft SQL Server Installation

- SQL Server Installation
 - SQL Server Configuration
 - SQL Server Services
 - SQL Server Management Studio (SSMS)
 - Instance Configuration
 - System Databases
 - TempDB Configuration
 - SQL Server Upgrade & Patching
-

03. Database Administration Essentials

- Database Creation
- Filegroups & Data Files
- Storage Management

- Tables & Indexes
 - Identity Columns
 - Database Maintenance
 - Import & Export Data
 - Bulk Insert & BCP Utility
-

04. SQL Server Security

- Authentication Modes
 - Logins & Users
 - Database Roles
 - Server Roles
 - Permissions & Privileges
 - Transparent Data Encryption (TDE)
 - Always Encrypted
 - Auditing
 - Row-Level Security
 - Dynamic Data Masking
-

05. Backup & Recovery

- Recovery Models
 - Full Backup
 - Differential Backup
 - Transaction Log Backup
 - Restore Database
 - Point-in-Time Recovery
 - Tail Log Backup
 - Backup Compression
 - Backup Encryption
 - Disaster Recovery Planning
-

06. Performance Tuning

- SQL Server Monitoring
- DMVs
- Wait Statistics
- Execution Plans
- Query Optimization

- Index Optimization
 - Missing Index Analysis
 - Statistics Management
 - Memory Optimization
 - Resource Governor
-

07. High Availability & Disaster Recovery

- SQL Replication
 - Log Shipping
 - Database Mirroring
 - Always On Availability Groups
 - Windows Failover Clustering
 - Failover Cluster Instances
 - Disaster Recovery Architecture
 - HA Best Practices
-

08. Azure SQL Administration

- Microsoft Azure Fundamentals
 - Azure Portal
 - Azure Resource Groups
 - Azure SQL Database
 - Azure SQL Managed Instance
 - Azure SQL Virtual Machine
 - Elastic Pools
 - Azure Storage
 - Azure Networking Basics
-

09. Azure Database Management

- Azure SQL Security
- Firewall Rules
- Private Endpoints
- Azure Backup
- Geo-Replication
- Failover Groups
- Azure Monitoring
- Azure Advisor

- Azure Defender for SQL
 - Scaling & Performance Optimization
-

10. Advanced DBA & Cloud Migration

- Azure Database Migration Service (DMS)
 - SQL Server Migration to Azure
 - Azure Data Studio
 - SQL Agent Jobs
 - PowerShell Automation
 - Azure Automation
 - Policy-Based Management
 - Database Health Checks
 - Capacity Planning
 - Enterprise DBA Best Practices
-



Hands-on Projects

- Install SQL Server from Scratch
 - Configure Enterprise SQL Environment
 - Design Production Database
 - User & Security Administration
 - Backup & Restore Strategy
 - Performance Optimization Case Study
 - Always On Availability Group Setup
 - SQL Replication Configuration
 - Migrate SQL Server to Azure
 - Azure SQL Production Deployment
-



Tools Covered

- Microsoft SQL Server
- SQL Server Management Studio (SSMS)
- Azure Portal
- Azure Data Studio
- Azure SQL Database
- Azure SQL Managed Instance
- PowerShell

- SQLCMD
 - Azure Monitor
 - Azure Database Migration Service
-



Career Opportunities

- Azure SQL Database Administrator
 - Microsoft SQL Server DBA
 - Database Administrator (DBA)
 - Azure Cloud Database Engineer
 - SQL Performance Engineer
 - Database Infrastructure Engineer
 - Azure Data Platform Engineer
 - Cloud Database Consultant
 - Database Operations Engineer
 - Senior SQL Server Administrator
-



Course Highlights

-  Beginner to Advanced Curriculum
-  Real-World Production Scenarios
-  Hands-on Lab Exercises
-  Enterprise DBA Best Practices
-  Backup & Disaster Recovery
-  Performance Tuning
-  Azure Cloud Administration
-  Migration to Azure
-  Interview Preparation
-  Job-Ready Skills

Tagline

"Master SQL Server & Azure Database Administration—Build, Secure, Optimize, and Manage Enterprise Databases with Confidence."